

# ⚡ Sort by Structure: Language Model Ranking as Dependency Probing

Max Müller-Eberstein<sup>✉</sup> and Rob van der Goot<sup>✉</sup> and Barbara Plank<sup>✉▲</sup>

<sup>✉</sup> Department of Computer Science, IT University of Copenhagen, Denmark

<sup>▲</sup> Center for Information and Language Processing (CIS), LMU Munich, Germany

mamy@itu.dk, robv@itu.dk, bplank@cis.lmu.de

## Abstract

Making an informed choice of pre-trained language model (LM) is critical for performance, yet environmentally costly, and as such widely underexplored. The field of Computer Vision has begun to tackle encoder ranking, with promising forays into Natural Language Processing, however they lack coverage of linguistic tasks such as structured prediction. We propose *probing to rank LMs*, specifically for parsing dependencies in a given language, by measuring the degree to which labeled trees are recoverable from an LM’s contextualized embeddings. Across 46 typologically and architecturally diverse LM-language pairs, our probing approach predicts the best LM choice 79% of the time using orders of magnitude less compute than training a full parser. Within this study, we identify and analyze one recently proposed decoupled LM—RemBERT—and find it strikingly contains less inherent dependency information, but often yields the best parser after full fine-tuning. Without this outlier our approach identifies the best LM in 89% of cases.

## 1 Introduction

With the advent of massively pre-trained language models (LMs) in Natural Language Processing (NLP), it has become crucial for practitioners to choose the best LM encoder for their given task early on, regardless of the rest of their proposed model architecture. The greatest variation of LMs lies in the language or domain-specificity of the unlabelled data used during pre-training (with architectures often staying identical).

Typically, better expressivity is expected from language/domain-specific LMs (Gururangan et al., 2020; Dai et al., 2020) while open-domain settings necessitate high-capacity models with access to as much pre-training data as possible. This tradeoff is difficult to navigate, and given that multiple specialized LMs (or none at all) are available, practitioners often resort to an ad-hoc choice. In absence of im-

mediate performance indicators, the most accurate choice could be made by training the full model using each LM candidate, however this is often infeasible and wasteful (Strubell et al., 2019).

Recently, the field of Computer Vision (CV) has attempted to tackle this problem by quantifying useful information in pre-trained image encoders as measured directly on labeled target data without fine-tuning (Nguyen et al., 2020; You et al., 2021). While first forays for applying these methods to NLP are promising, some linguistic tasks differ substantially: Structured prediction, such as parsing syntactic dependencies, is a fundamental NLP task not covered by prior encoder ranking methods due to its graphical output. Simultaneously, performance prediction in NLP has so far been studied as a function of dataset and model characteristics (Xia et al., 2020; Ye et al., 2021) and has yet to examine how to rank large pools of pre-trained LMs.

Given the closely related field of probing, in which lightweight models quantify task-specific information in pre-trained LMs, we recast its objective in the context of performance prediction and ask: *How predictive is lightweight probing at choosing the best performing LM for dependency parsing?* To answer this question, we contribute:

- An efficient encoder ranking method for structured prediction using dependency probing (Müller-Eberstein et al., 2022; DEPProbe) to quantify latent syntax (Section 2).
- Experiments across 46 typologically and architecturally diverse LM + target language combinations (Section 3).<sup>1</sup>
- An in-depth analysis of the surprisingly low inherent dependency information in RemBERT (Chung et al., 2021) compared to its high fine-tuned performance (Section 4).

<sup>1</sup>Code at <https://personads.me/x/naacl-2022-code>.

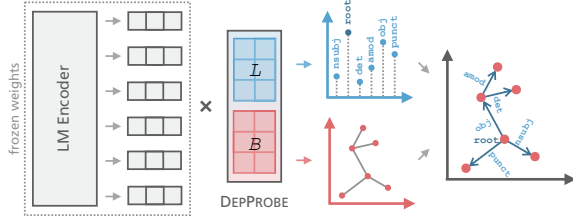


Figure 1: **Visualization of DEPProbe.** Relational and structural subspaces  $L$  and  $B$  are combined to extract labeled, directed trees from embeddings.

## 2 Methodology

Probing pre-trained LMs is highly related to encoder ranking in CV where the ease of recoverability of class-differentiating information is key (Nguyen et al., 2020; You et al., 2021). This approach is more immediate than existing NLP performance prediction methods which rely on featurized representations of source and target data without actively ranking encoders (Xia et al., 2020; Ye et al., 2021). As most experiments in NLP are conducted using a limited set of LMs—often a single model—without strong prior motivations, we see *LM ranking as a critical task on its own*.

While probes for LMs come in many forms, they are generally characterized as lightweight, minimal architectures intended to solve a particular task (Hall Maudslay et al., 2020). While non-linear models such as small multi-layer perceptrons are often used (Tenney et al., 2019), there have been criticisms given that their performance highly depends on the complexity of their architecture (Hewitt and Liang, 2019; Voita and Titov, 2020). As such, we rely on linear probes alone, which have the benefit of being extremely lightweight, closely resembling existing performance prediction methods (You et al., 2021), and allow for statements about linear subspaces contained in LM latent spaces.

**DEPProbe** (Müller-Eberstein et al., 2022; visualized in Figure 1) is a linear formulation for extracting fully labeled dependency trees based on the structural probe by Hewitt and Manning (2019). Given contextualized embeddings of dimensionality  $d$ , a linear transformation  $B \in \mathbb{R}^{b \times d}$  with  $b \ll d$  (typically  $b = 128$ ) maps them into a subspace in which the Euclidean distance between embeddings corresponds to the number of edges between the respective words in the gold dependency graph.

In our formulation, we supplement a linear transformation  $L \in \mathbb{R}^{l \times d}$  (with  $l$  = number of dependency relations) which maps each embedding to a

subspace in which the magnitude of each dimension corresponds to the likelihood of a word and its head being governed by a certain relation.

By computing the minimum spanning tree in  $B$  and then finding the word with the highest root likelihood in  $L$ , we can determine the directionality of all edges as pointing away from the root. All remaining edges are labeled according to the most likely non-root class in  $L$ , resulting in a fully directed and labeled dependency tree.

Note that this approach differs substantially from prior approaches which yield undirected and/or unlabeled trees (Hewitt and Manning, 2019; Kulmizev et al., 2020) or use pre-computed edges and non-linear classifiers (Tenney et al., 2019). DEPProbe efficiently computes the full target metric (i.e. labeled attachment scores) instead of approximate alternatives (e.g. undirected, unlabeled attachment scores or tree depth correlation).

## 3 Experiments

**Setup** We investigate the ability of DEPProbe to select the best performing LM for dependency parsing across nine linguistically diverse treebanks from Universal Dependencies (Zeman et al., 2021; UD) which were previously chosen by Smith et al. (2018) to reflect diverse writing systems and morphological complexity (see Appendix A).

For each target language, we employ three multilingual LMs—mBERT (Devlin et al., 2019), XLM-R (Conneau et al., 2020), RemBERT (Chung et al., 2021)—as well as 1–3 language-specific LMs retrieved by popularity from HuggingFace’s Model Hub (Wolf et al., 2020), resulting in a total of 46 LM-target pair setups (see Appendix C).

For each combination, we train a DEPProbe to compute labeled attachment scores (LAS), hypothesizing that LMs from which trees are most accurately recoverable also perform better in a fully tuned parser. To evaluate the true downstream performance of a fully-tuned model, we further train a deep biaffine attention parser (BAP; Dozat and Manning, 2017) on each LM-target combination. Compared to full fine-tuning, DEPProbe only optimizes the matrices  $B$  and  $L$ , resulting in the extraction of labeled trees with as few as 190k instead of 583M trainable parameters for the largest RemBERT model (details in Appendix B).

We measure the predictive power of probing for fully fine-tuned model performance using the Pearson correlation coefficient  $\rho$  as well as the weighted

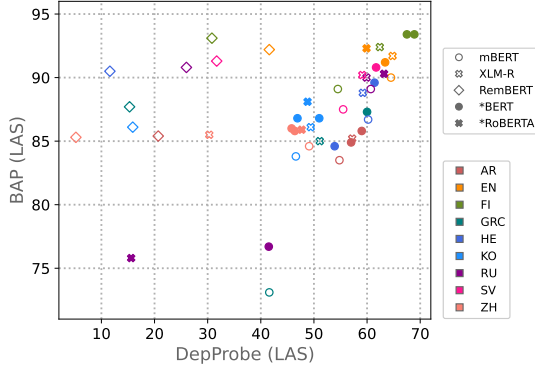


Figure 2: **LAS of DEPProbe in relation to full BAP** across nine language targets (dev) using language-specific and multilingual LM encoders of different architecture types (exact scores in Appendix C).

Kendall’s  $\tau_w$  (Vigna, 2015). The latter metric corresponds to a correlation coefficient in  $[-1, 1]$  and simultaneously defines the probability of choosing the better LM given a pair as  $\frac{\tau_w + 1}{2}$ , allowing us to quantify the overall quality of a ranking.

**Results** Comparing the LAS of DEPProbe’s lightweight predictions against full BAP fine-tuning in Figure 2, we see a clear correlation as the probe correctly predicts the difficulty of parsing languages relative to each other and also ranks models within languages closely according to their final performance. With a  $\tau_w$  of .58 between scores ( $p < 0.001$ ), this works out to DEPProbe selecting the better performing final model given any two models 79% of the time. Additionally, LAS is slightly more predictive of final performance than unlabeled, undirected attachment scores (UUAS) with  $\tau_w = .57$  to which prior probing approaches are restricted (see Appendix C).

Given a modest  $\rho$  of .32 ( $p < 0.05$ ), we surprisingly also observe a single strong outlier to this pattern, namely the multilingual RemBERT (Chung et al., 2021) decoupled LM architecture. While DEPProbe consistently ranks it low as it cannot extract dependency parse trees as accurately as from the BERT and RoBERTa-based architectures, RemBERT actually performs best on four out of the nine targets when fully fine-tuned in BAP. Excluding monolingual LMs, it further outperforms the other multilingual LMs in seven out of nine cases. As it is a more recent and distinctive architecture with many differences to the most commonly-used contemporary LMs, we analyze potential reasons for this discrepancy in Section 4.

Excluding RemBERT as an outlier, we find sub-

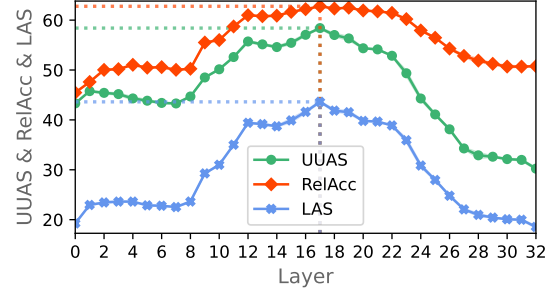


Figure 3: **Dependency Information per RemBERT Layer** via DEPProbe’s structural, relational and parsing accuracy (UUAS, RelAcc, LAS) on EN-EWT (dev).

stantially higher correlation among all other models:  $\rho = .78$  and  $\tau_w = .78$  ( $p < 0.001$ ). This means that among these models, fully fine-tuning the LM for which DEPProbe extracts the highest scores, yields the better final performance 89% of the time.

In practice, learning DEPProbe’s linear transformations while keeping the LM frozen is multiple orders of magnitude more efficient than fully training a complex parser plus the LM’s parameters. As such, linear probing offers a viable method for selecting the best encoder in absence of qualitative heuristics or intuitions. This predictive performance is furthermore achievable in minutes compared to hours and at a far lower energy budget (see Appendices B and C).

## 4 Probing Decoupled LMs

Considering DEPProbe’s high predictive performance across LMs with varying architecture types, languages/domains and pre-training procedures, we next investigate its limitations: Specifically, which differences in RemBERT (Chung et al., 2021) lead to it being measured as an outlier with seemingly low amounts of latent dependency information despite reaching some of the highest scores after full fine-tuning. The architecture has 32 layers and embeddings with  $d = 1152$ , compared to most models’ 12 layers and  $d = 768$ . It accommodates these size and depth increases within a manageable parameter envelope by using smaller input embeddings with  $d_{in} = 256$ . While choosing different  $d$  for the input and output embeddings is not possible in most prior models due to both embedding matrices being coupled, RemBERT decouples them, leading to a larger parameter budget and less overfitting on the masked language modeling pre-training task (Chung et al., 2021).

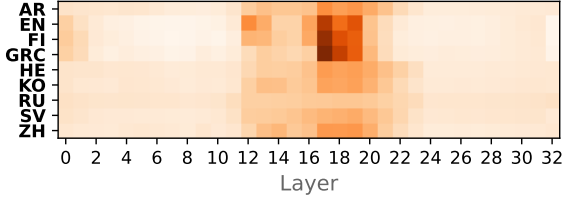


Figure 4: **Per-language  $\alpha$  of RemBERT Layers** for DEPProbe across all layer weights (dark > light).

| MODEL   | AR        | EN        | FI        | GRC       | HE        | KO        | RU        | SV        | ZH        |
|---------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| mBERT   | <b>65</b> | <b>74</b> | 65        | 46        | <b>69</b> | <b>58</b> | <b>68</b> | <b>65</b> | <b>58</b> |
|         | $\pm.08$  | $\pm.09$  | $\pm.35$  | $\pm.14$  | $\pm.23$  | $\pm.18$  | $\pm.31$  | $\pm.12$  | $\pm.17$  |
| XLM-R   | 60        | 70        | <b>66</b> | 53        | 60        | 49        | 57        | 51        | 51        |
|         | $\pm.14$  | $\pm.08$  | $\pm.18$  | $\pm.19$  | $\pm.20$  | $\pm.08$  | $\pm.34$  | $\pm.24$  | $\pm.53$  |
| RemBERT | 58        | 56        | 52        | <b>54</b> | 52        | 46        | 49        | 43        | 39        |
|         | $\pm.12$  | $\pm.22$  | $\pm.15$  | $\pm.18$  | $\pm.05$  | $\pm.14$  | $\pm.04$  | $\pm.08$  | $\pm.24$  |

Table 1: **LAS of BAP Trained on Frozen LMs.** A biaffine attention parsing head is trained on top of frozen mBERT, XLM-R and RemBERT for each of the nine target languages ( $\pm$  standard deviation).

**Layer-wise Probing** Prior probing studies have found dependency information to be concentrated around the middle layers of an LM (Hewitt and Manning, 2019; Tenney et al., 2019; Fayyaz et al., 2021). Using EN-EWT (Silveira et al., 2014), we evaluate whether this holds for RemBERT’s new architecture. Figure 3 confirms that both dependency structural and relational information are most prominent around layer 17 of 32 as indicated by UUAS and relation classification accuracy (RelAcc) respectively. Combining the structural and relational information in DEPProbe similarly leads to a peak of the LAS at the same layer while decreasing with further distance from the center.

Across all target languages, we next investigate whether probing a sum over the embeddings of all layers weighted by  $\alpha \in \mathbb{R}^{32}$  can boost extraction performance in RemBERT. The heavier weighting of middle layers by  $\alpha$ , visible in Figure 4, reaffirms a concentration of dependency information in the center. Contrasting probing work on prior models (Tenney et al., 2019; Kulmizev et al., 2020), using all layers does not increase the retrievable dependencies, with LAS differences  $\pm 1$  point. This further confirms that there is not a lack of dependency information in any specific layer, but that there is less within the encoder as a whole.

**Frozen Parsing** Our probing results show that linear subspaces in RemBERT contain less dependency information than prior LMs. However, DEPProbe’s parametrization is kept intentionally sim-

ple and may therefore not be capturing non-linearly represented information that is useful during later fine-tuning. To evaluate this hypothesis, we train a full biaffine attention parsing head, but keep the underlying LM encoder frozen. This allows us to quantify the performance gains which come from inherent dependency information versus later task-specific fine-tuning.

Table 1 confirms our findings from DEPProbe and shows that despite RemBERT outperforming mBERT and XLM-R when fully fine-tuned, it has substantially lower LAS across almost all languages when no full model fine-tuning is applied. This leads us to conclude that there indeed is less inherent dependency information in the newer model and that most performance gains must be occurring during task-specific full fine-tuning.

Given that DEPProbe extracts dependency structures reliably from LM architectures with different depths and embedding dimensionalities (e.g. RoBERTa<sub>large</sub> with 24 layers and  $d = 1024$  versus RuBERT<sub>tiny</sub> with 3 layers and  $d = 312$ ) as well as varying tokenization, optimization and pre-training data, the key difference in RemBERT appears to be embedding decoupling. The probe’s linear formulation is not the limiting factor as the non-linear, biaffine attention head also produces less accurate parses when the LM’s weights are frozen. Our analyses thus suggest that RemBERT’s decoupled architecture contains less dependency information out-of-the-box, but follows prior patterns such as consolidating dependency information towards its middle layers and serving as strong initialization for parser training.

Lastly, RemBERT’s larger number of tunable parameters compared to all other LM candidates may provide it further capacity, especially after full fine-tuning. As our probing methods are deliberately applied to the frozen representations of the encoder, it becomes especially important to consider the degree to which these embeddings may change after updating large parts of the model. Taking these limitations into account, the high correlations with respect to encoder ranking nonetheless enable a much more informed selection of LMs from a larger pool than was previously possible.

## 5 Conclusion

To guide practitioners in their choice of LM encoder for the structured prediction task of dependency parsing, we leveraged a lightweight, linear



DEPPROBE to quantify the latent syntactic information via the *labeled* attachment score. Evaluating 46 pairs of multilingual/language-specific LMs and nine typologically diverse target treebanks, we found DEPPROBE to not only be efficient in its predictions, with orders of magnitude fewer trainable parameters, but to also be accurate 79–89% of the time in predicting which LM will outperform another when used in a fully tuned parser. This allows for a substantially faster iteration over potential LM candidates, saving hours worth of compute in practice (Section 3).

Our experiments further revealed surprising insights on the newly proposed RemBERT architecture: While particularly effective for multilingual dependency parsing when fully fine-tuned, it contains substantially less latent dependency information relative to prior widely-used models such as mBERT and XLM-R. Among its architectural differences, we identified embedding decoupling to be the most likely contributor, while added model capacity during fine-tuning may also improve final performance. Our analyses showed that despite containing less dependency information overall, RemBERT follows prior findings such as structure and syntactic relations being consolidated towards the middle layers. Given these consistencies, performance differences between decoupled LMs may be predictable using probes, but in absence of similar multilingual LMs using decoupled embeddings this effect remains to be studied (Section 4).

Overall, the high efficiency and predictive power of ranking LM encoders via linear probing as well as the ease with which they can be analyzed—even when they encounter their limitations—offers immediate benefits to practitioners who have so far had to rely on their own intuitions when making a selection. This opens up avenues for future research by extending these methods to more tasks and LM architectures in order to enable better informed modeling decisions.

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## Appendices

### A Treebanks

| TARGET       | LANG          | FAMILY        | SIZE  |
|--------------|---------------|---------------|-------|
| AR-PADT      | Arabic        | Afro-Asiatic  | 7.6k  |
| EN-EWT       | English       | Indo-European | 16.6k |
| FI-TDT       | Finnish       | Uralic        | 15.1k |
| GRC-PROIEL   | Ancient Greek | Indo-European | 17.1k |
| HE-HTB       | Hebrew        | Afro-Asiatic  | 6.2k  |
| KO-GSD       | Korean        | Korean        | 6.3k  |
| RU-GSD       | Russian       | Indo-European | 5k    |
| SV-Talbanken | Swedish       | Indo-European | 6.0k  |
| ZH-GSD       | Chinese       | Sino-Tibetan  | 5.0k  |

Table 2: **Target Treebanks** based on [Smith et al. \(2018\)](#) with language family (FAMILY) and total number of sentences (SIZE).

Table 2 lists the nine target treebanks based on the set by [Smith et al. \(2018\)](#): AR-PADT ([Hajič et al., 2009](#)), EN-EWT ([Silveira et al., 2014](#)), FI-

TDT (Pyysalo et al., 2015), GRC-PROIEL (Eckhoff et al., 2018), HE-HTB (McDonald et al., 2013a), KO-GSD (Chun et al., 2018), RU-GSD (McDonald et al., 2013b), SV-Talbanken (McDonald et al., 2013a), ZH-GSD (Shen et al., 2016). We use these treebanks as provided in Universal Dependencies v2.9 (Zeman et al., 2021). DEPProbe and BAP are trained on each target’s respective training split and are evaluated on the development split as this work aims to analyze general performance patterns instead of state-of-the-art performance.

## B Experiment Setup

**DEPProbe** is implemented in PyTorch v1.9.0 (Paszke et al., 2019) and uses language models from the Transformers library v4.13.0 and the associated Model Hub (Wolf et al., 2020). Following the structural probe by Hewitt and Manning (2019), each token which is split by the LM encoder into multiple subwords is mean-pooled. Similarly, we follow the original hyperparameter settings and set the structural subspace dimensionality to  $b = 128$  and use embeddings from the middle layer of each LM (Hewitt and Manning, 2019; Tenney et al., 2019; Fayyaz et al., 2021). The structural loss is computed based on the absolute difference of the Euclidean distance between transformed word embeddings and the number of edges separating the words in the gold tree (see Hewitt and Manning, 2019 for details). The relational loss is computed using cross entropy between the logits and gold head-child relation. Optimization uses AdamW (Loshchilov and Hutter, 2018) with a learning rate of  $10^{-3}$  which is reduced by a factor of 10 each time the loss plateaus. Early stopping is applied after three epochs without improvement and a maximum of 30 total epochs. With the only trainable parameters being the matrices  $B$  and  $L$ , the model’s footprint ranges between 51k and 190k parameters.

**BAP** For the biaffine attention parser (Dozat and Manning, 2017) we use the implementation in the MaChAmp framework v0.3 (van der Goot et al., 2021) with the default training schedule and hyperparameters. The number of trainable parameters depends on the LM encoder’s size and ranges between 14M and 583M.

**Analyses** For our analyses in Sections 3 and 4 we further make use of numpy v1.21.0 (Harris et al., 2020), SciPy v1.7.0 (Virtanen et al., 2020) and Matplotlib v3.4.3 (Hunter, 2007).

**Training Details** Models are trained on an NVIDIA A100 GPU with 40GBs of VRAM and an AMD Epyc 7662 CPU. BAP requires around 1 h ( $\pm 30$  min). DEPProbe can be trained in around 15 min ( $\pm 5$  min) with the embedding forward operation being most computationally expensive. The models use batches of size 32 and are initialized using the random seeds 692, 710 and 932.

**Reproducibility** In order to ensure reproducibility and comparability with future work, we release our code and token-level predictions at <https://personads.me/x/naacl-2022-code>.

## C Detailed Results

Tables 3–11 list exact LAS and standard deviations for each experiment in Section 3’s Figure 2 in addition to the HuggingFace Model Hub IDs of the LMs used in each of the 46 setups as well as their number of layers, embedding dimensionality  $d$  and total number of parameters. In addition, Figure 5 shows UUAS for all setups, equivalent to only probing structurally (Hewitt and Manning, 2019) for unlabeled, undirected dependency trees.

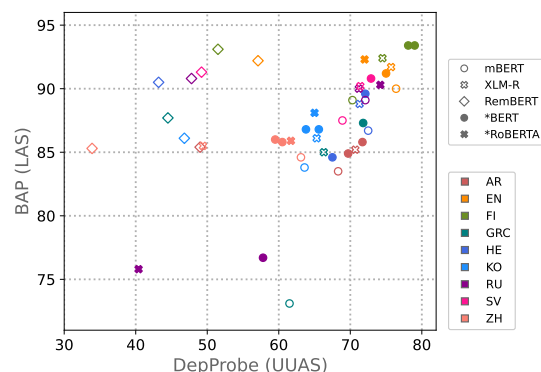


Figure 5: UUAS of DEPProbe in relation to BAP across nine language targets (dev) using language-specific and multilingual LM encoders of different architecture types.

| MODELS                          | SOURCE                | LAYERS | EMB $d$ | PARAMS | BAP            | DEPPROBE       |
|---------------------------------|-----------------------|--------|---------|--------|----------------|----------------|
| bert-base-multilingual-cased    | Devlin et al. (2019)  | 12     | 768     | 178M   | 83.5 $\pm$ 0.2 | 54.8 $\pm$ 0.6 |
| xlm-roberta-base                | Conneau et al. (2020) | 12     | 768     | 278M   | 85.2 $\pm$ 0.1 | 57.2 $\pm$ 0.1 |
| google/rembert                  | Chung et al. (2021)   | 32     | 1152    | 576M   | 85.4 $\pm$ 0.2 | 20.7 $\pm$ 0.1 |
| aubmindlab/bert-base-arabertv02 | Antoun et al. (2020)  | 12     | 768     | 135M   | 85.8 $\pm$ 0.1 | 59.0 $\pm$ 0.1 |
| asafaya/bert-base-arabic        | Safaya et al. (2020)  | 12     | 768     | 111M   | 84.9 $\pm$ 0.1 | 57.0 $\pm$ 0.2 |

Table 3: **LAS on AR-PADT (Dev)** using BAP and DEPPROBE with different LMs ( $\pm$  standard deviation).

| MODELS                       | SOURCE                | LAYERS | EMB $d$ | PARAMS | BAP            | DEPPROBE       |
|------------------------------|-----------------------|--------|---------|--------|----------------|----------------|
| bert-base-multilingual-cased | Devlin et al. (2019)  | 12     | 768     | 178M   | 90.0 $\pm$ 0.1 | 64.5 $\pm$ 0.3 |
| xlm-roberta-base             | Conneau et al. (2020) | 12     | 768     | 278M   | 91.7 $\pm$ 0.2 | 64.8 $\pm$ 0.1 |
| google/rembert               | Chung et al. (2021)   | 32     | 1152    | 576M   | 92.2 $\pm$ 0.0 | 41.6 $\pm$ 0.3 |
| bert-base-uncased            | Devlin et al. (2019)  | 12     | 768     | 109M   | 91.2 $\pm$ 0.1 | 63.4 $\pm$ 0.3 |
| roberta-large                | Liu et al. (2019)     | 24     | 1024    | 355M   | 92.3 $\pm$ 0.2 | 59.9 $\pm$ 0.2 |

Table 4: **LAS on EN-EWT (Dev)** using BAP and DEPPROBE with different LMs ( $\pm$  standard deviation).

| MODELS                                | SOURCE                 | LAYERS | EMB $d$ | PARAMS | BAP            | DEPPROBE       |
|---------------------------------------|------------------------|--------|---------|--------|----------------|----------------|
| bert-base-multilingual-cased          | Devlin et al. (2019)   | 12     | 768     | 178M   | 89.1 $\pm$ 0.2 | 54.5 $\pm$ 0.4 |
| xlm-roberta-base                      | Conneau et al. (2020)  | 12     | 768     | 278M   | 92.4 $\pm$ 0.1 | 62.4 $\pm$ 0.2 |
| google/rembert                        | Chung et al. (2021)    | 32     | 1152    | 576M   | 93.1 $\pm$ 0.1 | 30.8 $\pm$ 0.1 |
| TurkuNLP/bert-base-finnish-uncased-v1 | Virtanen et al. (2019) | 12     | 768     | 125M   | 93.4 $\pm$ 0.1 | 68.9 $\pm$ 0.3 |
| TurkuNLP/bert-base-finnish-cased-v1   | Virtanen et al. (2019) | 12     | 768     | 125M   | 93.4 $\pm$ 0.1 | 67.5 $\pm$ 0.4 |

Table 5: **LAS on FI-TDT (Dev)** using BAP and DEPPROBE with different LMs ( $\pm$  standard deviation).

| MODELS                             | SOURCE                    | LAYERS | EMB $d$ | PARAMS | BAP            | DEPPROBE       |
|------------------------------------|---------------------------|--------|---------|--------|----------------|----------------|
| bert-base-multilingual-cased       | Devlin et al. (2019)      | 12     | 768     | 178M   | 73.1 $\pm$ 0.1 | 41.6 $\pm$ 0.5 |
| xlm-roberta-base                   | Conneau et al. (2020)     | 12     | 768     | 278M   | 85.0 $\pm$ 0.2 | 51.1 $\pm$ 0.2 |
| google/rembert                     | Chung et al. (2021)       | 32     | 1152    | 576M   | 87.7 $\pm$ 0.1 | 15.3 $\pm$ 0.1 |
| pranaydeeps/Ancient-Greek-BERT     | Singh et al. (2021)       | 12     | 768     | 113M   | 87.3 $\pm$ 0.1 | 60.0 $\pm$ 0.0 |
| nlpaueb/bert-base-greek-uncased-v1 | Koutsikakis et al. (2020) | 12     | 768     | 113M   | 84.6 $\pm$ 0.3 | 53.9 $\pm$ 0.1 |

Table 6: **LAS on GRC-PROIEL (Dev)** using BAP and DEPPROBE with different LMs ( $\pm$  standard deviation).

| MODELS                       | SOURCE                | LAYERS | EMB $d$ | PARAMS | BAP            | DEPPROBE       |
|------------------------------|-----------------------|--------|---------|--------|----------------|----------------|
| bert-base-multilingual-cased | Devlin et al. (2019)  | 12     | 768     | 178M   | 86.7 $\pm$ 0.2 | 60.2 $\pm$ 0.6 |
| xlm-roberta-base             | Conneau et al. (2020) | 12     | 768     | 278M   | 88.8 $\pm$ 0.1 | 59.2 $\pm$ 0.3 |
| google/rembert               | Chung et al. (2021)   | 32     | 1152    | 576M   | 90.5 $\pm$ 0.1 | 11.6 $\pm$ 0.4 |
| onlplab/alephbert-base       | Seker et al. (2021)   | 12     | 768     | 126M   | 89.6 $\pm$ 0.1 | 61.4 $\pm$ 0.2 |

Table 7: **LAS on HE-HTB (Dev)** using BAP and DEPPROBE with different LMs ( $\pm$  standard deviation).

| MODELS                       | SOURCE                | LAYERS | EMB $d$ | PARAMS | BAP            | DEPPROBE       |
|------------------------------|-----------------------|--------|---------|--------|----------------|----------------|
| bert-base-multilingual-cased | Devlin et al. (2019)  | 12     | 768     | 178M   | 83.8 $\pm$ 0.2 | 46.6 $\pm$ 0.2 |
| xlm-roberta-base             | Conneau et al. (2020) | 12     | 768     | 278M   | 86.1 $\pm$ 0.1 | 49.4 $\pm$ 0.3 |
| google/rembert               | Chung et al. (2021)   | 32     | 1152    | 576M   | 86.1 $\pm$ 0.2 | 15.9 $\pm$ 0.3 |
| klue/bert-base               | Park et al. (2021)    | 12     | 768     | 111M   | 86.8 $\pm$ 0.0 | 51.0 $\pm$ 0.1 |
| klue/roberta-large           | Park et al. (2021)    | 24     | 1024    | 337M   | 88.1 $\pm$ 0.3 | 48.8 $\pm$ 0.5 |
| kykim/bert-kor-base          | Kim (2020)            | 12     | 768     | 118M   | 86.8 $\pm$ 0.1 | 46.9 $\pm$ 0.4 |

Table 8: **LAS on KO-GSD (Dev)** using BAP and DEPPROBE with different LMs ( $\pm$  standard deviation).



| MODELS                          | SOURCE                                | LAYERS | EMB $d$ | PARAMS | BAP            | DEPProbe       |
|---------------------------------|---------------------------------------|--------|---------|--------|----------------|----------------|
| bert-base-multilingual-cased    | <a href="#">Devlin et al. (2019)</a>  | 12     | 768     | 178M   | 89.1 $\pm$ 0.1 | 60.7 $\pm$ 0.1 |
| xlm-roberta-base                | <a href="#">Conneau et al. (2020)</a> | 12     | 768     | 278M   | 90.0 $\pm$ 0.2 | 59.9 $\pm$ 1.1 |
| google/rembert                  | <a href="#">Chung et al. (2021)</a>   | 32     | 1152    | 576M   | 90.8 $\pm$ 0.0 | 26.0 $\pm$ 0.2 |
| cointegrated/rubert-tiny        | <a href="#">Dale (2021)</a>           | 3      | 312     | 11M    | 76.7 $\pm$ 0.1 | 41.5 $\pm$ 0.6 |
| sberbank-ai/ruRoberta-large     | <a href="#">Sber Devices (2021)</a>   | 24     | 1024    | 355M   | 90.3 $\pm$ 0.3 | 63.2 $\pm$ 0.4 |
| blinoff/roberta-base-russian-v0 | <a href="#">Blinov (2021)</a>         | 12     | 768     | 124M   | 75.8 $\pm$ 0.0 | 15.6 $\pm$ 0.2 |

Table 9: **LAS on RU-GSD (Dev)** using BAP and DEPProbe with different LMs ( $\pm$  standard deviation).

| MODELS                       | SOURCE                                 | LAYERS | EMB $d$ | PARAMS | BAP            | DEPProbe       |
|------------------------------|----------------------------------------|--------|---------|--------|----------------|----------------|
| bert-base-multilingual-cased | <a href="#">Devlin et al. (2019)</a>   | 12     | 768     | 178M   | 87.5 $\pm$ 0.1 | 55.5 $\pm$ 0.2 |
| xlm-roberta-base             | <a href="#">Conneau et al. (2020)</a>  | 12     | 768     | 278M   | 90.2 $\pm$ 0.1 | 59.1 $\pm$ 0.2 |
| google/rembert               | <a href="#">Chung et al. (2021)</a>    | 32     | 1152    | 576M   | 91.3 $\pm$ 0.3 | 31.7 $\pm$ 0.3 |
| KB/bert-base-swedish-cased   | <a href="#">Malmsten et al. (2020)</a> | 12     | 768     | 125M   | 90.8 $\pm$ 0.1 | 61.7 $\pm$ 0.2 |

Table 10: **LAS on SV-Talbanken (Dev)** using BAP and DEPProbe with different LMs ( $\pm$  standard deviation).

| MODELS                       | SOURCE                                | LAYERS | EMB $d$ | PARAMS | BAP            | DEPProbe       |
|------------------------------|---------------------------------------|--------|---------|--------|----------------|----------------|
| bert-base-multilingual-cased | <a href="#">Devlin et al. (2019)</a>  | 12     | 768     | 178M   | 84.6 $\pm$ 0.4 | 49.1 $\pm$ 0.4 |
| xlm-roberta-base             | <a href="#">Conneau et al. (2020)</a> | 12     | 768     | 278M   | 85.5 $\pm$ 0.3 | 30.3 $\pm$ 0.1 |
| google/rembert               | <a href="#">Chung et al. (2021)</a>   | 32     | 1152    | 576M   | 85.3 $\pm$ 0.2 | 5.2 $\pm$ 0.1  |
| bert-base-chinese            | <a href="#">Devlin et al. (2019)</a>  | 12     | 768     | 102M   | 85.8 $\pm$ 0.1 | 46.4 $\pm$ 0.1 |
| hfl/chinese-bert-wwm-ext     | <a href="#">Cui et al. (2021)</a>     | 12     | 768     | 102M   | 86.0 $\pm$ 0.3 | 45.8 $\pm$ 0.3 |
| hfl/chinese-roberta-wwm-ext  | <a href="#">Cui et al. (2021)</a>     | 12     | 768     | 102M   | 85.9 $\pm$ 0.3 | 47.7 $\pm$ 0.4 |

Table 11: **LAS on ZH-GSD (Dev)** using BAP and DEPProbe with different LMs ( $\pm$  standard deviation).